EduPulse Analytics — Student Outcomes Analysis

Junior Data Analyst Case Study

Dataset: 20 student records supplied for this case study
Prepared for: EduPulse Analytics
Analysis scope: Marks, attendance, and pass outcomes

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1. Introduction

This report evaluates student outcomes in the supplied EduPulse Analytics dataset. It summarizes academic performance, tests the observed association between attendance and marks, applies a transparent passing rule, and presents outcome-focused visualizations. The analysis is descriptive: it identifies patterns in this sample but does not establish that attendance causes changes in marks.

2. Objectives

1. Calculate average student marks.
2. Measure and interpret the correlation between attendance and marks.
3. Create a `Pass` field using the rule **Marks ≥ 50**.
4. Summarize total passes and fails.
5. Visualize the distribution of marks.
6. Highlight high performers and students requiring support.

3. Dataset description

The dataset contains **20 students**, with one record per student.

Field	Description
`Student_ID`	Unique student identifier
`Name`	Student label/name

`Age`	Student age in years
`Marks`	Academic marks, out of 100
`Attendance_Percentage`	Attendance rate, expressed as a percentage
`Pass`	Derived Boolean outcome: `True` when marks are at least 50

No dates, course/subject labels, demographic attributes beyond age, or repeated assessment periods were provided. Findings therefore apply only to this 20-student snapshot.

4. Python analysis code

```
import pandas as pd
import matplotlib.pyplot as plt
import seaborn as sns
from scipy.stats import pearsonr

# Load the supplied dataset
df = pd.read_csv("student_performance_dataset.csv")

# 1. Average marks
average_marks = df["Marks"].mean()

# 2. Attendance–marks association
correlation, p_value = pearsonr(
    df["Attendance_Percentage"], df["Marks"]
)

# 3. Pass-status field
df["Pass"] = df["Marks"] >= 50

# 4. Pass/fail summary
pass_fail_summary = (
    df["Pass"]
      .map({True: "Pass", False: "Fail"})
      .value_counts()
      .rename_axis("Outcome")
      .reset_index(name="Student_Count")
)

# 5. Histogram of marks
sns.set_theme(style="whitegrid")
plt.figure(figsize=(8, 5))
sns.histplot(df, x="Marks", bins=8, color="#2A6FBB", edgecolor="white")
plt.title("Distribution of Student Marks")
plt.xlabel("Marks (out of 100)")
plt.ylabel("Number of Students")
plt.grid(True, alpha=0.3)
plt.tight_layout()
plt.show()
```

The report calculations used the provided 20-record dataset. The filename in the code is a portable placeholder; adjust it to the local file location when running independently.

5. Results and observations

5.1 Summary statistics

Measure	Result
Students analyzed	20
Average marks	**74.55 / 100**
Median marks	75.50 / 100
Marks standard deviation	18.17
Lowest marks	45 / 100
Highest marks	99 / 100
Average attendance	81.60%
Attendance range	61%–99%

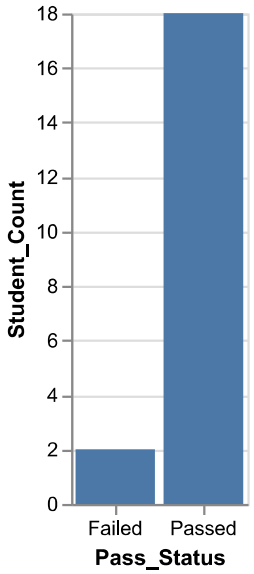
Observation: Average performance was above the passing threshold, although marks varied substantially across students (45 to 99).

5.2 Pass/fail outcome

The derived `Pass` column was set to `True` for marks of 50 or higher and `False` otherwise.

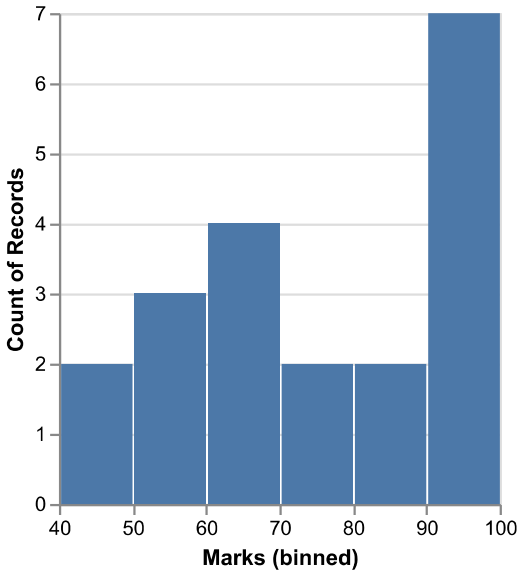
Outcome	Student count	Share of students
Pass (`True`)	**18**	**90.0%**

Fail (`False`)	**2**	**10.0%**
Total	**20**	**100.0%**



Observation: Eighteen students met the defined passing standard. Two students—**Student_12 (46 marks)** and **Student_18 (45 marks)**—were below it.

5.3 Distribution of student marks

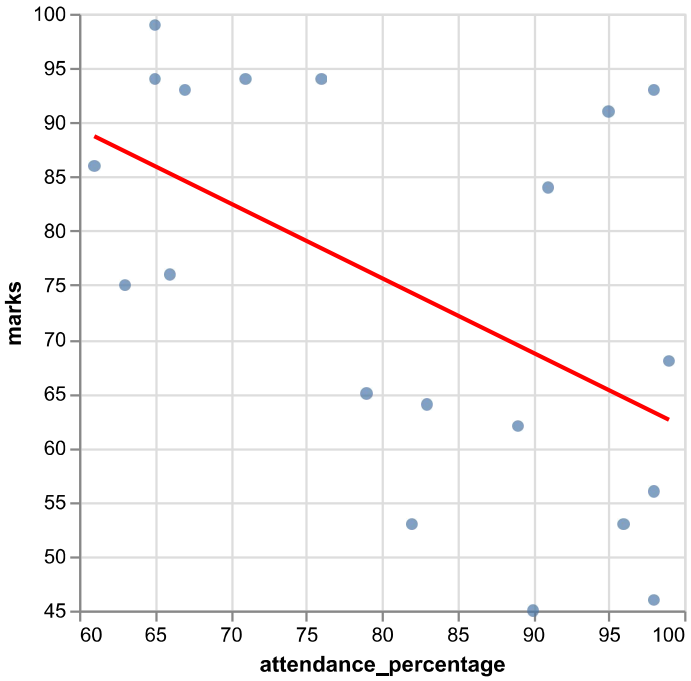


Observation: The histogram shows marks spanning a wide range, with the upper end of the distribution containing several strong outcomes. The sample includes two scores below 50 and a number of scores in the 90s.

5.4 Attendance and marks

Pearson correlation analysis was used to quantify the linear association between attendance percentage and marks.

Statistic	Result
Students analyzed	20
Pearson correlation coefficient, *r*	**−0.524**
Two-sided p-value	**0.018**

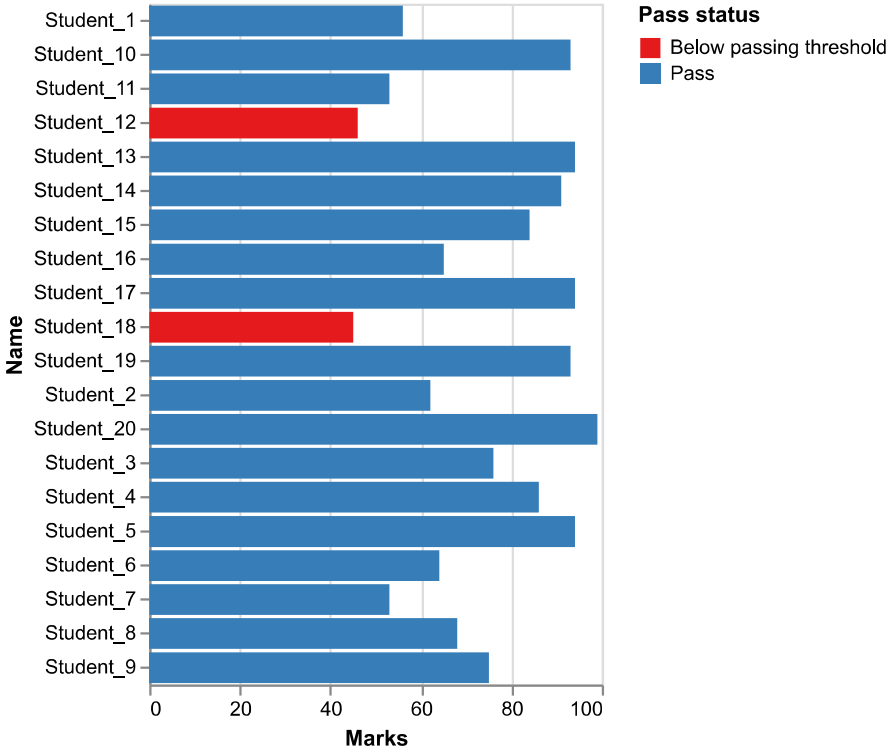


Interpretation: In this dataset, attendance and marks have a **moderate negative linear correlation** ($r = -0.524$). The p-value is below 0.05, indicating that this association would be unlikely under a zero-correlation model for this sample. This result is **not causal evidence** and should not be interpreted as attendance reducing marks. With only 20 observations and no controls for other factors, the relationship may reflect sample composition, confounding variables, or individual observations.

5.5 High-performing students

Rank	Student	Marks	Attendance

1	Student_20	99	65%
2	Student_5	94	71%
3	Student_13	94	76%
4	Student_17	94	65%
5	Student_10	93	98%



Observation: Student_20 achieved the highest mark (99). Four students shared or closely clustered at the next highest levels (93–94). The ranked view also makes the two below-threshold outcomes visible.

6. Key findings

- The cohort's average mark was **74.55**, which is **24.55 points above** the 50-mark passing threshold.
- **18 of 20 students (90.0%)** passed under the stated rule.
- Marks were dispersed, ranging from **45 to 99**, so the strong cohort average masks meaningful differences between students.

- The sample shows a statistically significant **moderate negative** attendance–marks correlation ($r = -0.524$, $p = 0.018$). It is an observed association, not a causal conclusion.
- **Student_12** and **Student_18** require outcome follow-up because they were the only students below the 50-mark threshold. Their attendance was 98% and 90%, respectively, showing that attendance alone does not explain these two outcomes in this data.

7. Recommendations

1. **Provide targeted academic diagnostic support** for Student_12 and Student_18, such as assessment review, tutoring, or subject-specific feedback. Their high attendance suggests that support should look beyond attendance barriers.
2. **Preserve and study effective performance practices** among high-mark students, while avoiding assumptions that a single factor drove their results.
3. **Do not use attendance as a stand-alone predictor of achievement** based on this sample. Add subject, assessment difficulty, prior attainment, learning-support, and longitudinal data before creating an intervention rule.
4. **Monitor outcomes over additional assessment cycles.** A larger, repeated dataset would enable more reliable trend analysis and better identification of persistent risk.
5. **Maintain the explicit pass rule** (`Marks >= 50`) in the reporting workflow so outcome counts are reproducible and auditable.

8. Limitations

- The analysis contains only **20 students**, limiting statistical precision and generalizability.
- The data are cross-sectional; no time sequence is available to assess change or causality.
- Correlation is sensitive to the sample and does not control for subject, prior performance, instructional factors, or other confounders.
- Student names are labels in the supplied data; no additional contextual information is available for individual-level explanations.

9. Conclusion

The supplied cohort shows strong overall outcomes: the average mark was **74.55**, and **90%** of students passed. However, the mark range and the two below-threshold students indicate a need for focused academic follow-up rather than reliance on cohort averages. The observed negative attendance–marks correlation should be investigated with a larger and richer dataset before it informs policy or intervention decisions.